

Lia Chavez

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PROJECTS

AI Captain (RF-DETR Integration)

Spring 2026

Python, RF-DETR, COCO Labeling

- Integrated a Receptive Field-based Detection Transformer (RF-DETR) into an existing autonomous maritime navigation system to enhance real-time vessel detection and situational awareness.
- Addressed limitations of YOLOv8 and U-Net by developing an improved object-detection pipeline for identifying vessels across diverse marine environments.
- Managed and standardized COCO-format dataset annotations to support RF-DETR training and optimize vessel detection performance.
- Achieved 87% precision and 90% recall, demonstrating reliable vessel detection while minimizing false positives and missed detections.

Fashion Trend Forecasting

Sept 2025 – Present

Python, Machine Learning

- Developing a machine learning model to predict emerging trends in the fashion industry by analyzing historical data.

CRKL "Circle" Space

Sept 2025 – Present

PostgreSQL

- Developing a website to increase user productivity of navigating astronomy and space exploration related information by 50%.
- Designing and implementing a relational database.
- Leading team meetings and keeping teammates on track as team lead.

Autonomous Line Tracking Robot <https://www.youtube.com/watch?v=TrSsChyW2ws>

Fall 2025

C++

- Engineered an Arduino UNO R3-based robot capable of following a defined path using Infrared Sensors.
- Implemented real-time analog sensor processing for directional control and smooth path tracking.
- Designed and tested the full electrical system, performing iterative debugging for reliable performance.

EXPERIENCE

Embedded Systems Instructional Student Assistant

Aug 2025 – Dec 2025

California State University, San Marcos

San Marcos, CA

- Evaluated student assignments for an Embedded Systems Computer Science course.
- Supported students by clarifying concepts related to Embedded Systems.
- Maintained course documentation and ensured grading consistency.

SKILLS

Languages: Python, SQL, C/C++

Frameworks & Tools: Jupyter Notebook, Git/GitHub, Arduino IDE

Concepts: Machine Learning, Computer Vision, Databases, Embedded Systems

EDUCATION

San Diego State University

San Diego, CA

M.S. in Computer Science.

Expected Spring 2028

California State University, San Marcos

San Marcos, CA

B.S. in Computer Science. Dean's List 2022-2026. GPA: 3.7.

May 2026